Huawei Access Node BAS-ACC-07 BGP Route Churn and OSPF Adjacency Loss Root Cause Analysis and Resolution SOP

Node: BAS-ACC-07 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_7311

Description

The Huawei Access Node BAS-ACC-07, located at the Basavanagudi POP, is experiencing a Routing Protocol Flap failure, characterized by BGP route churn and OSPF adjacency loss. This failure is attributed to the complex interplay of multiple factors, including the dynamic nature of BGP and OSPF routing protocols, as well as the hardware mechanics of the Huawei Access Node. Specifically, the failure of the OSPF adjacency is likely due to the loss of synchronization between the OSPF LSAs (Link-State Advertisements) and the OSPF LSPs (Link-State Packets) on the Huawei Access Node. Similarly, the BGP route churn is likely caused by the dynamic nature of BGP routing, which involves the constant exchange of routing information between the Huawei Access Node and its BGP neighbors. The severity of this failure is classified as Major, with an alarm code of ALARM_7311.

Pre-requisite Checks

- prerequisite: Verify that the Basavanagudi POP is accessible via remote hands or on-site personnel
- prerequisite: Check the status of the redundant link to ensure that it is online and functioning correctly
- prerequisite: Review the system logs to determine the exact time and date of the failure

Potential Root Causes

Cause	Probability
OSPF adjacency loss due to LSA/LSP desynchronization	High
BGP route churn due to dynamic routing	High
Hardware failure or misconfiguration on the Huawei Access Node	Medium

Diagnostic Commands & Expected Outputs

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Command: display ospf peer, Expected Output: OSPF Peer Information, Description: This command displays information about the OSPF peer, including the peer ID, state, and LSA/LSR count. This information can help identify any issues with the OSPF adjacency.

Command: display bgp routing-table, Expected Output: BGP Routing Table, Description: This command displays information about the BGP routing table, including the prefix, next hop, and AS path. This information can help identify any issues with the BGP route churn.

Command: display huawei acl, Expected Output: Huawei ACL Configuration, Description: This command displays information about
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the Huawei ACL configuration, including the ACL ID, name, and rule.
This information can help identify any issues with the ACL
configuration.
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Resolution Steps

Step 1: Step: Clear OSPF adjacency loss, Command: clear ospf peer , Description: This command clears the OSPF adjacency loss by resetting the OSPF peer. If the OSPF adjacency is cleared, the OSPF LSA/LSP desynchronization may be resolved.

Step 2: Step: Reset BGP route churn, Command: reset bgp , Description: This command resets the BGP route churn by restarting the BGP session with the specified neighbor. If the BGP route churn is reset, the BGP dynamic routing may be resolved.

Step 3: Step: Verify Huawei ACL configuration, Command: display huawei acl , Description: This command verifies the Huawei ACL configuration to ensure that it is correct and not contributing to the failure.

Post-Check Verification

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Command: display ospf peer, Expected Output: OSPF Peer Information,
Description: This command verifies that the OSPF adjacency has been
cleared and that the OSPF LSA/LSP desynchronization has been resolved.

Command: display bgp routing-table, Expected Output: BGP Routing
Table, Description: This command verifies that the BGP route churn has
been reset and that the BGP dynamic routing has been resolved.
```

Roll-back Procedures

- Step: Restore original OSPF configuration, Command: undo clear ospf peer , Description: If the OSPF adjacency loss is not resolved, this command restores the original OSPF configuration by resetting the OSPF peer to its previous state.

- Step: Restore original BGP configuration, Command: undo reset bgp , Description: If the BGP route churn is not resolved, this command restores the original BGP configuration by restarting the BGP session with the specified neighbor to its previous state.